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Then, the Bayh-Dole Act allowed universities and non-profit institutions to retain title to patents derived from federally-funded research and development. The process continued with reform of the national laboratories in the US in 1986 when the government allowed them to enter into cooperative R&D arrangements with industry.

The story of any nascent renaissance of Indian public sector science must begin with the under-sung leadership of Raghunath Mashelkar, CSIR's one time head. Forty-two national laboratories started a major transformation programme in 1996 under his tutelage. The 'CSIR 2001 Vision Document', published in January 1996, outlined ambitious goals for the future including realising 10 per cent of operational expenditure from intellectual property licensing (up from less than 1 per cent prior to 1996) and earning \$40 million annually from overseas R&D (up from less than \$2 million). To achieve these goals, CSIR set out to work with one of the biggest multi-

Collective ideology:
Private entrepreneurs can serve as catalysts to reignite enthusiasm in moribund parts of the Indian state

national innovators around the block—General Electric (GE). In a great example of the private sector stirring up research in government labs, around 1991, CSIR's Pune-based National Chemical Laboratory (NCL) started interacting with GE, the latter being a large purchaser of a chemical named THPE. Hoechst Celanese, another multinational company, was the only supplier of THPE to the global market. In 1994, NCL initiated a programme, funded by GE aimed at developing a proprietary process for THPE. The GE-NCL alliance was successful in breaking the global monopoly of Hoechst in the area of THPE with NCL earning revenues of around \$8.5 million over a nine-year period from GE.

In addition to revenue, collaborating with the private sector can help dismantle the walls separating Indian government labs and universities. Researchers reporting to different government agencies have traditionally worked in their own silos. Unlike the US, there are few organisations like the National